

Student Observer: _____

Period: **PERIOD 0 HONORS**

Date: _____

OBSERVER PROTOCOL & SCIENTIFIC DEFENSE CRITERIA

1. Apparatus & Applied Impulse: Record the object/mass used, surface interface, and how the sudden force was applied to minimize impulse time (Δt) and friction.

2. Newton's 1st Law Defense: Explain why the mass behaved as observed using $\Sigma F = 0$. **CRITICAL CHECK:** Verify presenter does **NOT** claim an unphysical "forward force of inertia" pushed the mass!

3. Defense Score (1–4 pts):
4: Flawless 1st Law defense, zero forward force.
3: Minor clarity gap. **2:** Claimed inertia force.
1: Incomplete or unphysical defense.

DEMO & PRESENTER Name & Demo Title	CATEGORY Regime (Check 1)	APPARATUS & APPLIED IMPULSE Mass, support surface & how quick impulse was applied (Δt)	OBSERVED MOTION & PHYSICS DEFENSE ($\Sigma F = 0$) What mass did, Newton's 1st Law defense & check for forward force	SCORE 1–4 pts
DEMO 1 Presenter: _____ Codename: _____	☐ Rest ☐ Motion ☐ Direction		 Forward Force Claimed? ☐ Yes (Error) ☐ No (Accurate)	____ / 4 [4][3] [2][1]
DEMO 2 Presenter: _____ Codename: _____	☐ Rest ☐ Motion ☐ Direction		 Forward Force Claimed? ☐ Yes (Error) ☐ No (Accurate)	____ / 4 [4][3] [2][1]
DEMO 3 Presenter: _____ Codename: _____	☐ Rest ☐ Motion ☐ Direction		 Forward Force Claimed? ☐ Yes (Error) ☐ No (Accurate)	____ / 4 [4][3] [2][1]
DEMO 4 Presenter: _____ Codename: _____	☐ Rest ☐ Motion ☐ Direction		 Forward Force Claimed? ☐ Yes (Error) ☐ No (Accurate)	____ / 4 [4][3] [2][1]
DEMO 5 Presenter: _____ Codename: _____	☐ Rest ☐ Motion ☐ Direction		 Forward Force Claimed? ☐ Yes (Error) ☐ No (Accurate)	____ / 4 [4][3] [2][1]

UNIT 2: DYNAMICS · DEMONSTRATIONS 6–10 & MECHANICAL EQUILIBRIUM SYNTHESIS

DEMO & PRESENTER Name & Demo Title	CATEGORY Regime (Check 1)	APPARATUS & APPLIED IMPULSE Mass, support surface & how quick impulse was applied (Δt)	OBSERVED MOTION & PHYSICS DEFENSE ($\Sigma F = 0$) What mass did, Newton's 1st Law defense & check for forward force	SCORE 1–4 pts
DEMO 6 Presenter: _____ Codename: _____	☐ Rest ☐ Motion ☐ Direction	_____ _____ _____	_____ _____ _____ Forward Force Claimed? ☐ Yes (Error) ☐ No (Accurate)	____ / 4 [4][3] [2][1]
DEMO 7 Presenter: _____ Codename: _____	☐ Rest ☐ Motion ☐ Direction	_____ _____ _____	_____ _____ _____ Forward Force Claimed? ☐ Yes (Error) ☐ No (Accurate)	____ / 4 [4][3] [2][1]
DEMO 8 Presenter: _____ Codename: _____	☐ Rest ☐ Motion ☐ Direction	_____ _____ _____	_____ _____ _____ Forward Force Claimed? ☐ Yes (Error) ☐ No (Accurate)	____ / 4 [4][3] [2][1]
DEMO 9 Presenter: _____ Codename: _____	☐ Rest ☐ Motion ☐ Direction	_____ _____ _____	_____ _____ _____ Forward Force Claimed? ☐ Yes (Error) ☐ No (Accurate)	____ / 4 [4][3] [2][1]
DEMO 10 Presenter: _____ Codename: _____	☐ Rest ☐ Motion ☐ Direction	_____ _____ _____	_____ _____ _____ Forward Force Claimed? ☐ Yes (Error) ☐ No (Accurate)	____ / 4 [4][3] [2][1]

 HONORS SYNTHESIS: BRIDGING INERTIA TO BALANCED FORCES ($\Sigma F = 0$)

1. Forensic Analysis of Confounding Forces: Which demonstration in class achieved the cleanest isolation of inertia? Explain how the presenter minimized contact time (Δt) and friction to prevent unwanted force transfer:

2. The Equilibrium Condition: In every demonstration observed today, whenever the mass stayed at rest or moved at constant velocity, what was the horizontal net force acting on that mass? Write the equation: $\Sigma F_x =$ _____ N. What would occur if an unbalanced net force acted?

HONORS LOG DEFENSE SCORE:

• Observation Table (16 pts)

• Synthesis & Equilibrium Bridge (4 pts)

TOTAL: ____ / 20 pts

Teacher Signature: _____